SULIT

UTM Fakulti Komputeran
UNIVERSITI TEKNOLOGI MALAYSIA

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER 1, 2023/2024

SUBJECT CODE	:	**SECD2523**
SUBJECT NAME	:	**DATABASE**
SECTION	:	**01, 02, 03, 05, A06, 07, 08, 09, 10**
TIME	:	
DATE/DAY	:	
VENUES	:	

INSTRUCTIONS:

i. Answer **ALL** questions in the given **ANSWER BOOKLET**.
ii. This paper consists of **THREE SECTIONS:**
    Section A: 20 MCQs
    Section B: 4 Structured Questions
    Section C: 2 Case Study Questions
iii. Submit **BOTH Questions and Answer Booklet** at the end of the exam.

(Please Write Your Lecturer Name and Section in Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This question paper consists of **SEVENTEEN (17)** printed pages excluding this page.

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SECTION A

OBJECTIVE QUESTIONS (20 MARKS)

INSTRUCTION:

This section consists of **TWENTY (20)** questions. Answer **ALL** questions in the answer booklet.

1. In creating an entity, there are few rules that should be considered **EXCEPT**:

   A. Remove relationship name from entity name
   B. Name an entity using reserved words
   C. Give the entity a unique name
   D. Add a few attributes, if possible

2. Which one of the following is the best example of the composite attribute?

   A. MatricID
   B. ProductNo
   C. Address
   D. Email

3. The relationship between superclass/subclass is _____.

   A. one to one
   B. one to many
   C. zero to one
   D. many to many

4. How many table(s)/relation(s) required for constraints of specialization/generalization **{mandatory, and}** in EERD?

   A. More than 3
   B. 3
   C. 2
   D. 1

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5. *"All of the publications must be either a journal, or article or news".*

   What are the constraints on specialization / generalization based on the above sentence?

   A. {Mandatory, AND}
   B. {Mandatory, OR}
   C. {Optional, AND}
   D. {Optional, OR}

6. In derivation of a new relation in logical database design, entity on 1 side will be designated as _____, while entity on many (*) side will be designated as _____.

   A. parent, child
   B. child, parent
   C. parent, parent
   D. child, child

7. The following anomalies are the categories of update anomalies **EXCEPT**:

   A. Insertion anomalies
   B. Deletion anomalies
   C. Modification anomalies
   D. Selection anomalies

8. There are two types of design in data warehouse which are star schema and snowflake schema. Which of the following explains the **CORRECT** comparison between these types.

   A. Star – keeping the same number of systems but migrating each one to a larger system.
      Snowflake – when the workload exceeds server capacity, it is spread out across a number of servers
   B. Snowflake – when the workload exceeds server capacity, it is spread out across a number of servers.
      Snowflake – keeping the same number of systems but migrating each one to a larger system

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C. Star – when the workload exceeds server capacity, it is spread out across a number of servers.

Snowflake – A dimensional data model that has a fact table in the center, surrounded by normalized dimension tables.

D. Snowflake – keeping the same number of systems but migrating each one to a larger system.

Snowflake – A dimensional data model that has a fact table in the center, surrounded by denormalized dimension tables.

9. "**WHERE**" clause in SQL is used to _____.

A. Specify which columns to test for equality when two tables are joined.
B. Extract only those records that fulfil a specified condition.
C. Combine rows from two or more tables, based on a related column between them.
D. Add, delete, or modify columns in an existing table.

10. The following joins are the types of outer join **EXCEPT**:

A. Left join
B. Self join
C. Right join
D. Full join

11. Identify the best business rule for the following ERD in **Figure 1**.

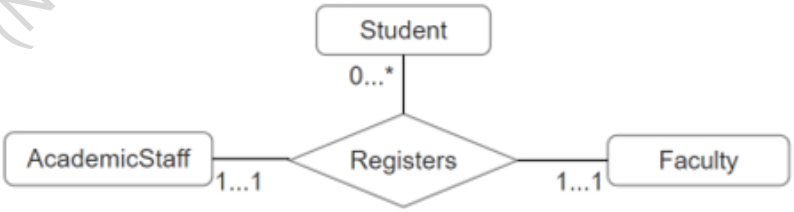


Figure 1

A. Academic staff of a faculty registers zero student or more.
B. Academic staff of a faculty registers many students.
C. Faculty will register one academic staff and many students.
D. Faculty will register one academic staff and zero student or more.

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12. Based on the following EERD in **Figure 2** and the given situations, determine the answer for a, b and c.

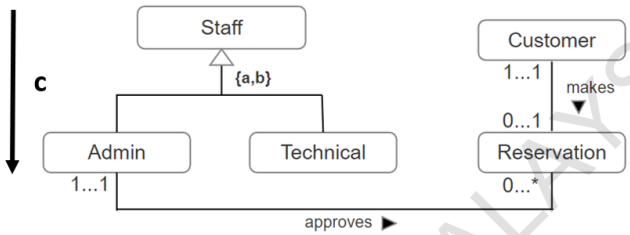


Figure 2

"25% of the staff in Company XYZ are the admins and 20% are the technical staff. The technical staff will only be focusing on their technical task."

A. a = mandatory, b = or, c = generalization
B. a = mandatory, b = and, c = specialization
C. a = optional, b = and, c = generalization
D. a = optional, b = or, c = specialization

13. Choose the **CORRECT** statement (s) that describes the following ERD as in **Figure 3**.

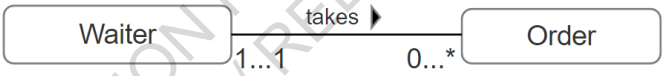


Figure 3

I. A waiter can take zero or many orders.
II. Each order will be taken by many waiters.
III. Waiter participation is optional.
IV. Order participation is optional.

A. I only
B. II and IV
C. I and III
D. II and III

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14. Which of the following statement is / are **FALSE** based on the given EERD as in **Figure 4**?

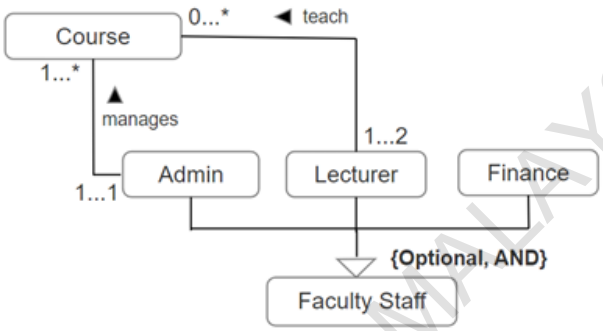


Figure 4

I. Staffs at faculty cannot be an admin and finance staff at the same time.

II. An admin can manage one or many courses.

III. Each lecturer can teach one to two courses only.

IV. A faculty staff also can be a research management staff.

A. I only

B. I and III

C. III and IV

D. I, II and IV

15. Which one is the **CORRECT** derivation of relation for the Local Logical Data Model based on the following relationship?

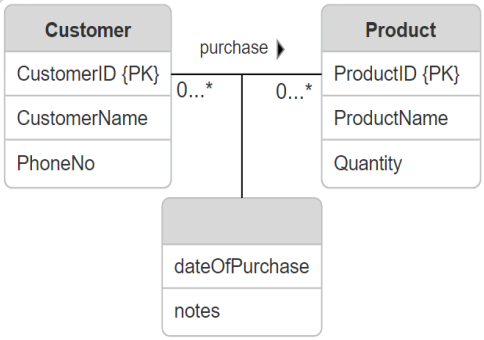


Figure 5

A. Customer (CustomerID, CustomerName, PhoneNo)
Product (ProductID, ProductName, Quantity)

B. Customer (CustomerID, CustomerName, PhoneNo, dateOfPurchase, notes)
Product (ProductID, ProductName, Quantity)

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C. Customer (CustomerID, CustomerName, PhoneNo)
Product (ProductID, ProductName, Quantity)
Purchase (CustomerID, dateOfPurchase, notes)

D. Customer (CustomerID, CustomerName, PhoneNo)
Product (ProductID, ProductName, Quantity)
Purchase (dateOfPurchase, notes, CustomerID, ProductID)

16. Describe the order of rules correctly for deriving relations that represent one-to-one (1:1) binary relationship type for optional participation on both sides.

I. Choose the entity which is closer to being optional as the parent.
II. Create a relation for each entity.
III. Copy the primary key attribute of parent entity into relation representing child entity, to act as a foreign key.
IV. Choose the entity which is closer to being mandatory as the child

A. I, IV, II, III
B. I, IV, III, II
C. II, IV, III, I
D. II, III, I, IV

17. **Table 1** shows the invoice data of customers that buy some books at Book Store Muhibbah.

Table 1: Invoice of data customers

CustID	CustName	Address	BookID	BookTitle	BookCategory	PricePerUnit	Quantity
101	Iznirah	Jalan Melati	501	Introduction to Machine Learning	Education	79.90	5
			404	Keluarga Bahagia	General	29.50	5
			607	Planets	Science	65.90	6
228	Bobby	Razak Street	803	Let's learn ABC	Kids	10.50	10
			404	Keluarga Bahagia	General	29.50	3

Choose the **CORRECT** relations that can be formed with its maximum level of normal form.

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A. Customer (CustID, CustName, Address)
Book (BookID, BookTitle, BookCategory, PricePerUnit, Quantity)

B. Customer (CustID, CustName, Address)
Book (BookID, BookTitle, BookCategory, PricePerUnit, Quantity)

C. Invoice (CustID, ProductID, Quantity)
Customer (CustID, CustName, Address)
Book (BookID, BookTitle, BookCategory, PricePerUnit)

D. Invoice (CustID, ProductID, Quantity)
Customer (CustID, CustName, Address)
Book (BookID, BookTitle, BookCategory, PricePerUnit)

18. Compare the characteristic of these two approaches of NoSQL database.

Bucket = Customer

Key	Value
10010	"LName Collin FName Alfred Initial A Areacode 615 Phone 844-2573 Balance 0"
10011	"LName Dunne FName Leona Initial K Areacode 713 Phone 894-1238 Balance 0"
10014	"LName Orlando FName Myron Areacode 615 Phone 222-1672 Balance 0"

Approach 1

Collection = Customer

Key	Document
10010	{LName: "Collin", FName: "Alfred", Initial: "A", Areacode: "615", Phone: "844-2573", Balance: "0"}
10011	{LName: "Dunne", FName: "Leona", Initial: "K", Areacode: "713", Phone: "894-1238", Balance: "0"}
10014	{LName: "Orlando", FName: "Myron", Areacode: "615", Phone: "222-1672", Balance: "0"}

Approach 2

A. **Approach 1**: Store data on relationship-rich data as a collection of nodes and edges.
Approach 2: Store data in key-value pairs in which the value components are encoded documents grouped into large groups called collections.

B. **Approach 1:** Store data in key-value pairs in which the value components are encoded documents grouped into large groups called collections.
Approach 2: Store data on relationship-rich data as a collection of nodes and edges.

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C. **Approach 1:** Store data as a collection of key-value pairs organized as buckets which are the equivalent of tables.
Approach 2: Store data in key-value pairs in which the value components are encoded documents grouped into large groups called collections.

D. **Approach 1:** Store data in key-value pairs in which the value components are encoded documents grouped into large groups called collections.
Approach 2: Store data as a collection of key-value pairs organized as buckets which are the equivalent of tables.

19. Maria wants to display all the athlete's data in the following Table 2 based on their height in descending order. Which query can perform this task?

Table 2: Athlete Relation

ID	Name	Sport	Age	Height
555	Zana Alia	Badminton	20	165
556	Wan Basri	Football	19	170
557	Peter William	Taekwando	21	175
558	Lisa Zuryani	Netball	20	158

A. SELECT * FROM Athlete ORDER BY Height DSC;
B. SELECT * FROM Athlete ORDER BY Height DESC;
C. SELECT Height FROM Athlete ORDER BY DSC;
D. SELECT Height FROM Athlete ORDER BY DESC;

20.

```
SELECT *
FROM Nurse INNER JOIN Patient
ON Nurse.PatientID = Patient.PatientID;
```

Choose the **CORRECT** description of the above query.

A. The query will retrieve all columns from the combined result set of the "Nurse" and "Patient" tables where the "PatientID" values match.
B. The query will retrieve PatientID and NurseID from the combined result set of the "Nurse" and "Patient" tables where the "PatientID" values match.
C. The query will retrieve all columns from the combined result set of the "Nurse" and "Patient" tables where the "PatientID" values does not match.

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D. The query will retrieve PatientID and NurseID from the combined result set of the "Nurse" and "Patient" tables where the "PatientID" values does not match.

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SECTION B

STRUCTURED QUESTIONS (50 MARKS)

INSTRUCTION:

This section consists of **FOUR (4)** questions. Answer **ALL** questions in the answer booklet.

QUESTION 1 **[7.5 MARKS]**

You are tasked with creating an inventory management system for a food vendor that operates multiple food stalls in different locations. The food stall vendor offers a variety of food items, each with its own set of ingredients. All food stall vendors will have multiple food items stored. Each of the food item will have ID, name of the food, type of food, price and available quantity.

The food stall vendor needs a database to track the inventory levels of both raw ingredients and finished food items. Ingredients such as flour, onions, garlic and anchovies can be shared among different food items, and the stall vendor wants to ensure efficient tracking of ingredient usage across all food items. The ingredient can be stored by having ID of each ingredient, name of ingredient, available quantity and unit of measure. The food item can have multiple ingredients, and an ingredient can be used in multiple food items.

Each food stall vendor has its own inventory, and they have updated sales of food items in its inventory. Each food stall will have ID, locations and inventory capacity stored in the database. Each stall can have many sales stored, and sales are recorded for only one stall. The food stall vendor also wants to monitor and track sales of food items. Hence, a sale can include multiple food items, and a food item can be sold in multiple sales. Each sale can be recorded by having SaleID, Date, Time and TotalAmount.

Based on the case study given, **DRAW** the complete entity-relationship model for the food vendor's inventory management system

(7.5 Marks)

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QUESTION 2 **[7.5 MARKS]**

For all systems in UTM, a user may be defined as Student, Staff or both, but need not be either of them. Information about the user includes user number, first name, last name and address. Each Student must be either a taught or a research student. Information on Student includes matric number and email. Meanwhile, Staff consist of information such as email and department. Staff can be one to three supervisors that supervise each research student.

Based on the description, **DRAW the EERD** and include relevant attributes of each superclass, subclasses, relationship and constraints using the UML notation.

(7.5 MARKS)

QUESTION 3 **[15 MARKS]**

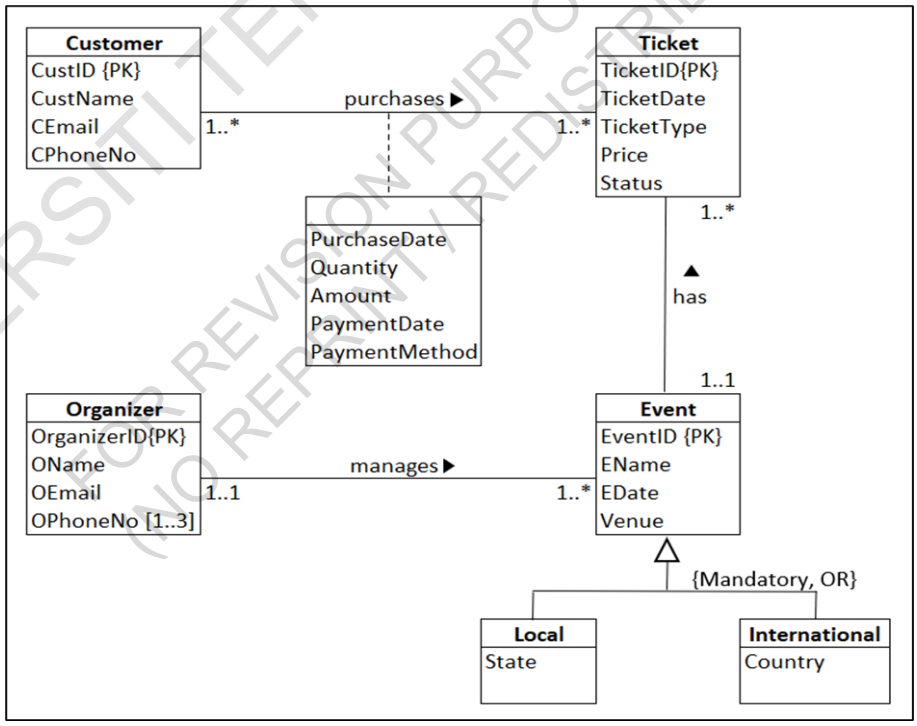


Figure 6: Conceptual ERD for Event Ticketing System

a) Transform the conceptual ERD in Figure 6 into a logical ERD.

(7 Marks)

b) Based on your answer in (a), derive all relations schema. State the primary key and foreign key (if any) for each relation schema.

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(8 Marks)

QUESTION 4 **[20 MARKS]**

The Traffic Safety Office at New Millenia College maintains a list of traffic offence tickets issued to students who violate traffic rules (such as vehicles parked illegally, driving exceeding the on-campus speed limit or blocking the traffic) on the campus. **Figure 7** shows an example of traffic records of a student that were kept in the Traffic Safety Office.

NEW MILLENIA COLLEGE
TRAFFIC SAFETY OFFICE

MATRIC NO SM203333 ACADEMIC ADVISOR (AA) ID L303
NAME Imran Fadly bin Muiz AA NAME Ng Looi Wen
PROGRAM BIT FACULTY NAME Fac of Technology

TICKET NO	DATE	FAULT CODE	FAULT NAME	FINE (RM)	FINE DUE	OFFICER ID	OFFICER NAME
21008	09/05/2021	P	Wrong parking area	50	31/12/2021	G200	SALEH MAN
21075	11/11/2021	P	Wrong parking area	50	31/12/2021	G405	ALI HARUN
22011	09/08/2022	S	Exceed speed limit	30	31/12/2022	G200	SALEH MAN
23455	13/06/2023	P	Wrong parking area	50	31/12/2023	G765	ROSMAN ABDUL
23707	02/12/2023	B	Blocking traffic	20	31/12/2023	G888	NOR AIN ALI

Figure 7: An Example of Traffic Records of a Student

As a database designer, you are asked to help the Public Safety Office create a better database design in order to eliminate data redundancy. In your first step, you identified ***Matric_No***, ***AcaAd_ID*** and ***Ticket_No*** as the primary keys, and you have produced the dependencies as below:

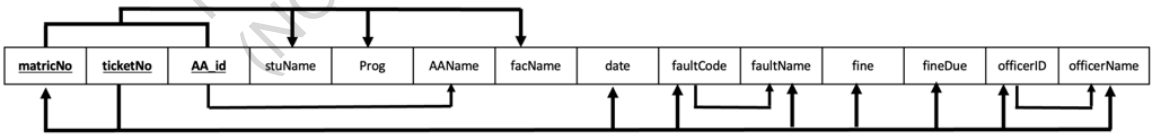


Figure 8

Answer question (a) – (d) to help you design the database for the Public Safety Office.

a) Based on **Figure 8**, identify the functional dependencies for each relation.

(5 Marks)

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b) Based on your answer in (a), remove the partial dependencies and produce relations schema in Second Normal Form (2NF).

(3 Marks)

c) Based on your answer in (b), remove the transitive dependencies and produce relations schema in the Third Normal Form (3NF).

(5 Marks)

d) Illustrate the tables based on your answer in (c) with the **data as shown in Figure 7.**

(7 Marks)

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SECTION C

CASE STUDY QUESTIONS (30 MARKS)

INSTRUCTION:

This section consists of **TWO (2)** questions. Answer **ALL** question in the answer booklet.

QUESTION 1 **[15 MARKS]**

This database scenario represents a hospital system where patients are scheduled for doctor appointments. Doctors have specialties, and patients may receive medication prescriptions based on their appointments. These relations allow tracking of patient details, doctor information, scheduled appointments, prescribed medications, and dosage instructions within the hospital system.

1. **Patient Relation**
   - Primary Key: patient_id

patient_id	patient_name	date_of_birth	patient_gender	patient_age
101	Sofea	2002-02-25	Female	21
102	Farah	2002-11-18	Female	22
103	Megat	2000-12-20	Male	23
104	Danial	2004-08-27	Male	19
105	Ammar	2002-02-13	Male	21

2. **Doctors Relation**
   - Primary Key: doctor_id

doctor_id	doctor_name	doctor_speciality
201	Dr. Idham	Cardiology
202	Dr. Balqis	Pediatrics
203	Dr. Adam	Orthopaedics
204	Dr. Umayrah	Neurology
205	Dr. Syuhadah	Oncology
206	Dr Thia	Pediatrics

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3. **Appointments Relation**
    - Primary Key: appt_id
    - Foreign Keys: patient_id (references Patient relation), doctor_id (references Doctor relation)

appt_id	patient_id	doctor_id	appt_date	appt_reason
1	101	201	2023-01-10 09:00:00	Checkup
2	103	202	2023-02-05 14:30:00	Vaccination
3	101	203	2023-03-05 11:30:00	Checkup
4	102	204	2023-04-20 10:00:00	MRI
5	104	205	2023-05-15 13:45:00	Chemotherapy

4. **Medication Relation**
    - Primary Key: med_id

med_id	med_name	med_dosage
101	Paracetamol	500mg
102	Amoxicillin	250mg
103	Ibuprofen	200mg
104	Aspirin	300mg
105	Insulin	10units

5. **Prescription Relation**
    - Primary Key: presc_id
    - Foreign Keys: patient_id (references Patients relation), med_id (references Medication relation)

presc_id	patient_id	med_id	presc_date	dosage_instructions
1	101	101	2023-01-11 10:00:00	As needed
2	102	101	2023-02-20 12:15:00	Once daily
3	103	103	2023-03-08 09:30:00	Twice daily
4	104	104	2023-04-05 14:45:00	After meals
5	105	105	2023-05-10 08:00:00	Daily

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a) Write the SQL statement that retrieves only the names and date of birth of all patients.

(1 Mark)

b) Write the SQL statement that lists all doctors with their specialty.

(1 Mark)

c) Display the output from the SQL statement below:

```
SELECT patient_name, date_of_birth
FROM Patients
WHERE patient_gender = 'Female';
```

(2 Marks)

d) Write the SQL statement that display all the appointments details for patient ID 101.

(2 Marks)

e) Write the SQL statement that displays all doctors who specialize in Pediatrics based on the table below:

doctor_id	doctor_name	doctor_speciality
202	Dr. Balqis	Pediatrics
206	Dr Thia	Pediatrics

(2 Marks)

f) Write the SQL statement that retrieves the names, dates of birth and ages of patients who are 21 years old.

(2 Marks)

g) Write the SQL statement that lists all the appointment details where the appointment date is between the month of January and February.

(2 Marks)

h) Write the SQL statement that display the prescription date and dosage instruction for medication id 101. Use Prescription Date and Prescription Dosage Instruction from relation Prescription as the column aliases.

(3 Marks)

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QUESTION 2 **[15 MARKS]**

Ahmad is working with an online retail database that includes tables for customers, orders, products, and product categories. The schema of the database is listed below.

Customer (customerID, fName, lName, email, country)

- Primary Key: customerID

Orders (orderID, customerID, orderDate, totalAmount)

- Primary Key: orderID
- Foreign Key: customerID references Customer(customerID)

Products (productID, productName, categoryID, price)

- Primary Key: productID
- Foreign Key: categoryID references ProductCategories(categoryID)

ProductCategories (categoryID, categoryName)

- Primary Key: categoryID

Write/construct the SQL statements for the queries below:

a) A list of customerID, fName, lName, email, and country sorted by their last name in descending order.

(2 Marks)

b) A list of orders containing orderID, customerID, and orderDate sorted by orderDate in descending order and customerID in ascending order.

(2 Marks)

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c) A list of the top 5 orders with the highest total amounts. The output should have Ranking, orderID, customerID, orderDate, and totalAmount. The output should be as shown in the table shown below:

Ranking	orderID	customerID	orderDate	totalAmount
1	D004523	C0023	2023-11-10	850.00
2	D002234	C0034	2023-08-31	800.00
3	D001123	C0123	2023-07-11	780.00
4	D003322	C0456	2023-10-11	750.00
5	D004455	C0876	2023-12-03	700.00

(5 Marks)

d) A list of orderID with customer details (customer ID, first name, last name, and email) for each order, join the tables using the ON clause.

(3 Marks)

e) A list of product names along with their category name. Include products with no assigned categories.

(3 Marks)

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-END OF QUESTIONS PAPER-